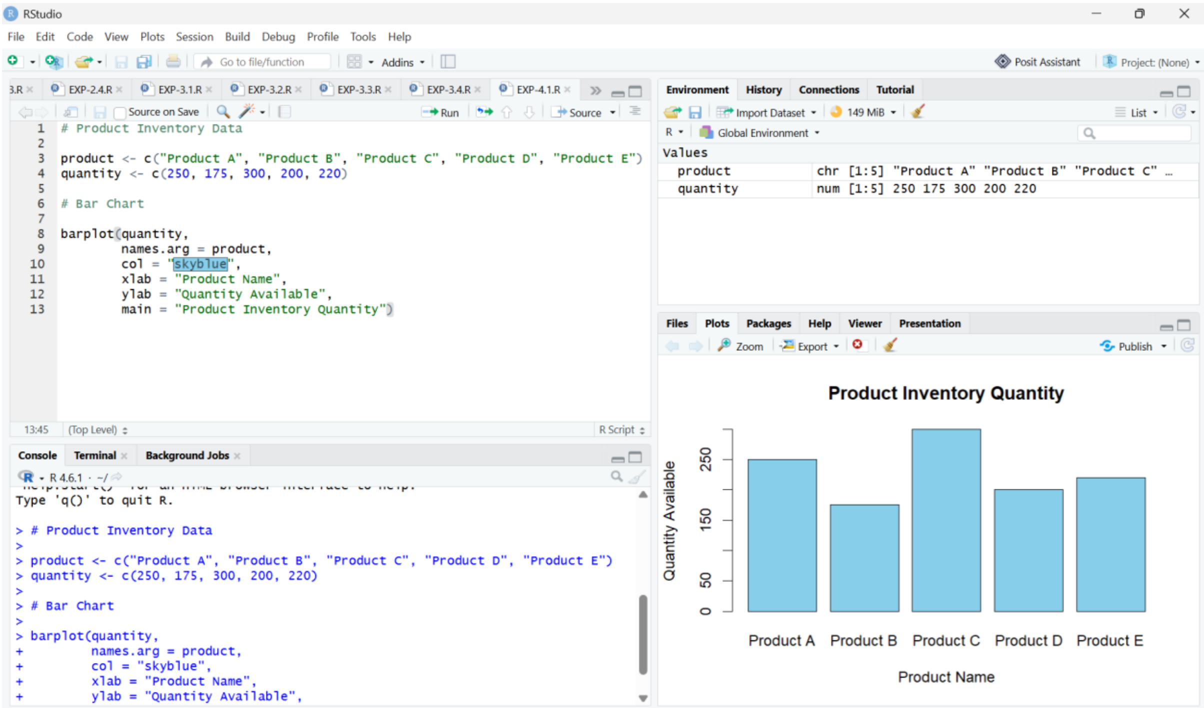
SET-4

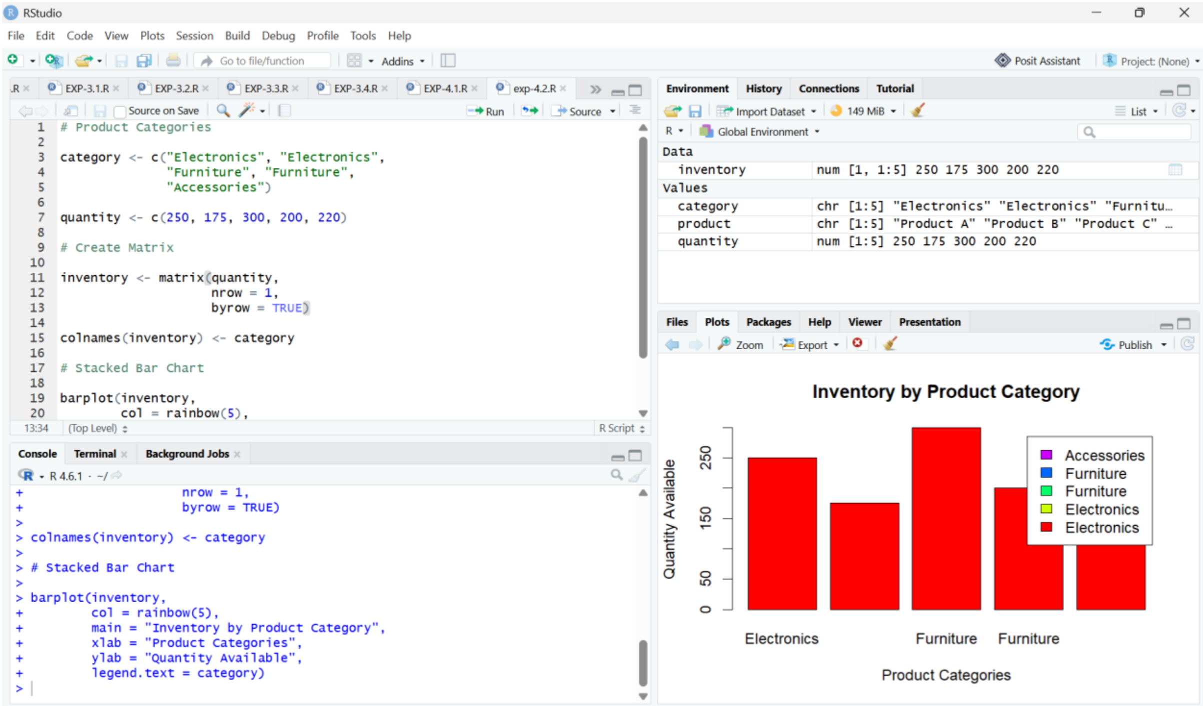
QUESTION-1



CODE:

```
# Product Inventory Data

product <- c("Product A", "Product B", "Product C", "Product D", "Product E")

quantity <- c(250, 175, 300, 200, 220)

# Bar Chart

barplot(quantity,

        names.arg = product,

        col = "skyblue",

        xlab = "Product Name",

        ylab = "Quantity Available",

        main = "Product Inventory Quantity")
```

QUESTION-2

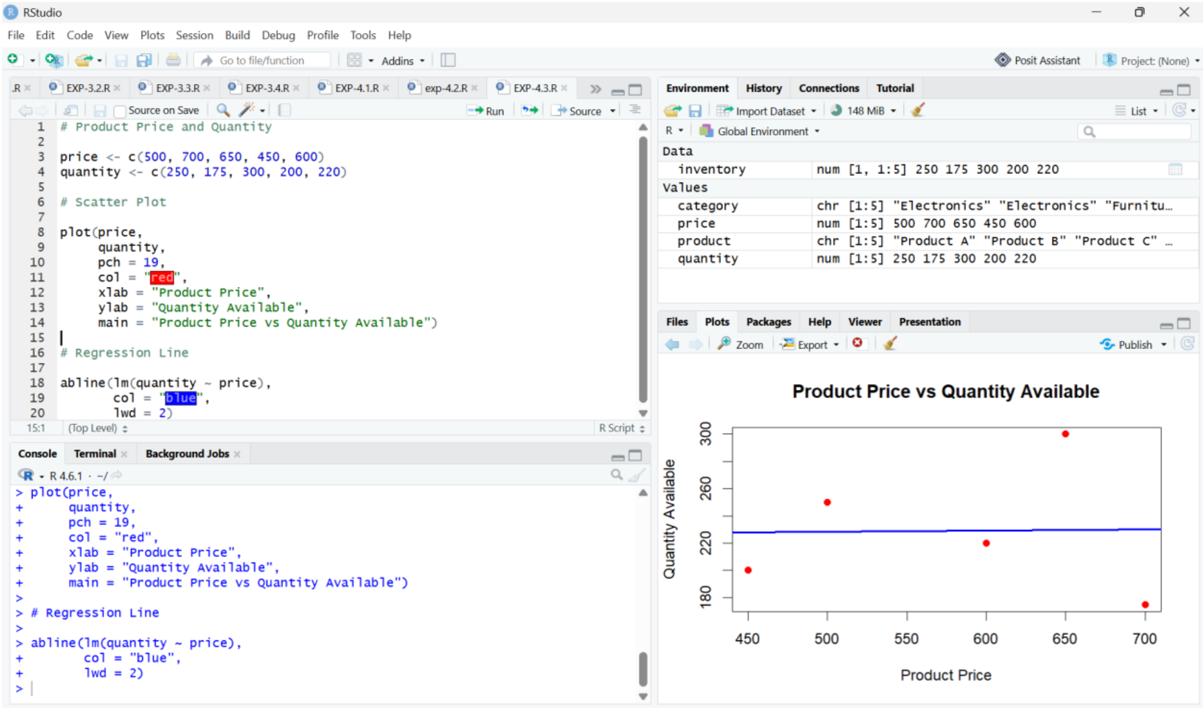


CODE:

```
# Product Categories

category <- c("Electronics", "Electronics",

              "Furniture", "Furniture",

              "Accessories")

quantity <- c(250, 175, 300, 200, 220)

# Create Matrix

inventory <- matrix(quantity,

                    nrow = 1,

                    byrow = TRUE)

colnames(inventory) <- category

# Stacked Bar Chart

barplot(inventory,

        col = rainbow(5),

        main = "Inventory by Product Category",

        xlab = "Product Categories",
```

```
    ylab = "Quantity Available",
    legend.text = category)
```

QUESTION-3



CODE:

```
# Product Price and Quantity


price <- c(500, 700, 650, 450, 600)

quantity <- c(250, 175, 300, 200, 220)


# Scatter Plot


plot(price,
     quantity,
     pch = 19,
     col = "red",
     xlab = "Product Price",
```

ylab = "Quantity Available",

main = "Product Price vs Quantity Available")

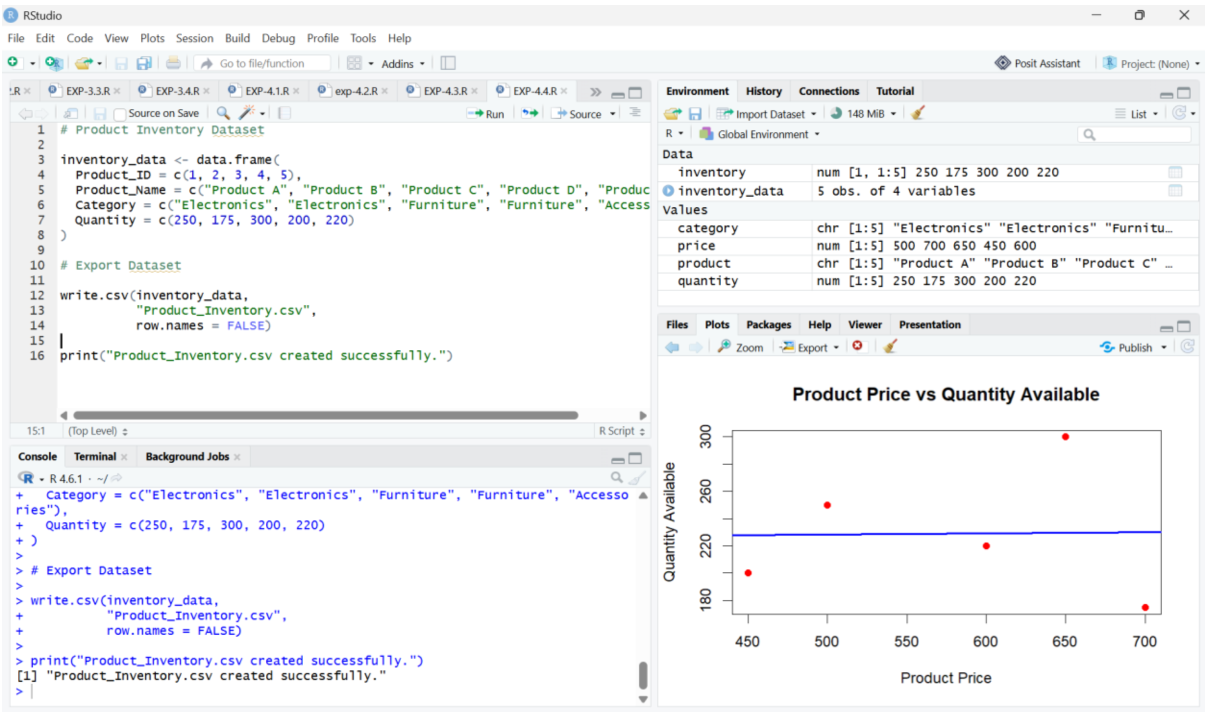
Regression Line

abline(lm(quantity ~ price),

col = "blue",

lwd = 2)

QUESTION-4



CODE:

Product Inventory Dataset

inventory_data <- data.frame(

Product_ID = c(1, 2, 3, 4, 5),

Product_Name = c("Product A", "Product B", "Product C", "Product D", "Product E"),

Category = c("Electronics", "Electronics", "Furniture", "Furniture", "Accessories"),

Quantity = c(250, 175, 300, 200, 220)

```
)

# Export Dataset
write.csv(inventory_data,
          "Product_Inventory.csv",
          row.names = FALSE)
print("Product_Inventory.csv created successfully.")
```